

ECON-7102: Environmental Economics I

Class 3: Theory of Externalities/Public Goods

Casey J. Wichman

School of Economics
Georgia Institute of Technology

Fall 2026

Today's Agenda

1. Questions on syllabus / course logistics?
2. Theory of externalities & public goods
3. For next class...

Pareto optimality (refresher)

An economic outcome is said to be Pareto optimal if a reallocation of resources cannot make at least one person better off without making another person worse off.

(Phaneuf & Requate, 2017, def'n 1.1)

- Focuses on maximizing the size of the pie.
- Does not consider distribution of benefits.

When Are Markets Efficient?

First Fundamental Theorem of Welfare Economics (FWT)

When markets are complete, any competitive equilibrium is necessarily Pareto optimal.

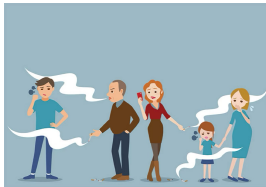
Second Fundamental Theorem of Welfare Economics (SWT)

Pareto optimal allocations can be supported as a price equilibrium with transfers (FWT + convex preferences).

Externalities

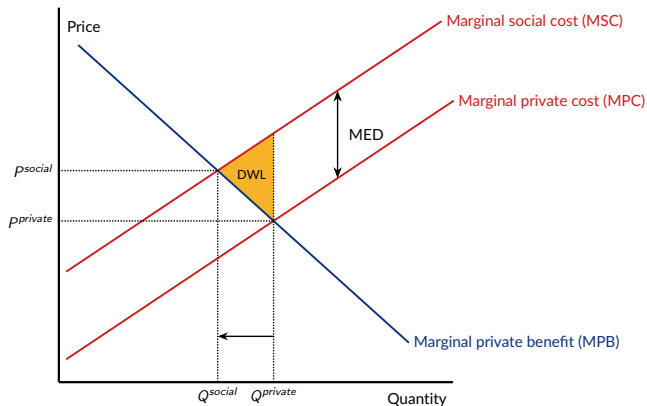
An externality exists when agent A's utility or production function depends directly on real variables chosen by another agent B, without an offer of compensation or other attention given to the effect on A's well-being.

(Phaneuf & Requate, def'n 1.2)



Externalities (an intermediate micro approach)

- Diff. Between MPC and MSC is **marginal external cost or damage (MED)**.
- $MPC + MED = MSC$
- Externalities lead to overproduction of dirty good.
- **DWL** is welfare loss to society because pollution goes unpriced.
- Policies seek to set incentives to align $Q^{private}$ and Q^{social}



2-person, 2-good economy

- Two individuals: $i = 1, 2$
 - Consume goods
 - Provide labor inputs
- Clean good, z_i , produced with labor inputs: $z = g(l_z)$
- Dirty good, x_i , produced with labor and emissions: $x = f(l_x, E)$
 - g, f are increasing in inputs (e.g., $\partial f / \partial E > 0$)
- Labor constraint: $l = l_x + l_z$
- Individuals gain utility over goods and emissions: $U_i(x_i, z_i, E)$

Pareto optimality, formally

- Maximize person 1's utility without making person 2 worse off, subject to technology/factor endowment constraints.

$$\begin{aligned} \max_{x_i, z_i, l_i, E} \quad & U_1(x_1, z_1, E) \\ & + \lambda_u [U_2(x_2, z_2, E) - \bar{u}_2] \quad (\text{P.O. condition}) \\ & + \lambda_x [f(l_x, E) - x_1 - x_2] \quad (\text{technology constraint for } x) \\ & + \lambda_z [g(l_z) - z_1 - z_2] \quad (\text{technology constraint for } z) \\ & + \lambda_l [l - l_x - l_z] \quad (\text{labor endowment constraint}) \end{aligned}$$

Pareto optimality, formally

Efficiency in consumption

$$\frac{\partial U_1}{\partial x_1} = \lambda_x, \quad \frac{\partial U_1}{\partial z_1} = \lambda_z, \quad \lambda_u \frac{\partial U_2}{\partial x_2} = \lambda_x, \quad \lambda_u \frac{\partial U_2}{\partial z_2} = \lambda_z$$

Combining...

$$\frac{\partial U_1 / \partial x_1}{\partial U_1 / \partial z_1} = \frac{\lambda_x}{\lambda_z} = \frac{\partial U_2 / \partial x_2}{\partial U_2 / \partial z_2}$$

$\Rightarrow$ **Marginal rate of substitution** (x for z) must be equal for both individuals.

Pareto optimality, formally

Efficiency in **production**

FOC w.r.t. labor:

$$\lambda_x \frac{\partial f}{\partial l_x} = \lambda_l, \quad \lambda_z \frac{\partial g}{\partial l_z} = \lambda_l$$

$\Rightarrow$ **Shadow price** of marginal product of labor for each good should equal shadow price of labor.

(Note: a **shadow price** is the value of relaxing a constraint by a small amount.)

Efficiency in exchange

Combining first-order conditions for consumption and production leads to:

$$\frac{\partial U_i / \partial x_i}{\partial U_i / \partial z_i} = \frac{\lambda_x}{\lambda_z} = \frac{\partial g / \partial l_z}{\partial f / \partial l_x} \quad \text{for } i = 1, 2$$

- $\implies$ for an allocation $\{x_1^*, x_2^*, z_1^*, z_2^*, l_x^*, l_z^*, E^*\}$ to be Pareto optimal, the slope of the PPF must be equal to the slope of each person's indifference curve.

Pareto optimality, formally

Efficiency in emissions

$$\frac{\partial U_1}{\partial E} + \lambda_u \frac{\partial U_2}{\partial E} + \lambda_x \frac{\partial f}{\partial E} = 0$$

Divide by λ_x and note that $\lambda_x = \partial U_1 / \partial x_1$ and $\lambda_u \partial U_2 / \partial x_2 = \lambda_x$, so:

$$\underbrace{-\frac{\partial U_1 / \partial E}{\partial U_1 / \partial x_1} - \frac{\partial U_2 / \partial E}{\partial U_2 / \partial x_2}}_{\text{Marginal willingness to give up } x \text{ for fewer emissions for both individuals.}} = \underbrace{\frac{\partial f}{\partial E}}_{\text{Physical marginal cost of reducing } E \text{ (in units of } x\text{)}}$$

- Although $\downarrow E$ increases utility directly, it decreases utility indirectly because there is less of x to consume.
- $\implies E^*$ needs to balance sum of marginal willingness to tradeoff E for x with marginal cost of reducing E .

Pareto optimality, summary

Recap

- Fancy way of saying that efficient consumption/production occurs where **indifference curves** and **production possibility curves** are tangent.
 - Think of an intermediate micro Edgeworth Box.
 - https://www.econgraphs.org/graphs/exchange/edgeworth_box/edgeworth_box_framework_v1
- Optimal emissions?
 - Set **MB** of emissions reduction to **MC** of reducing emissions.
- Relationship to public goods theory: optimal emissions condition on previous slide is essentially the same as Samuelson (1954) condition.
- Pareto optimality is a theoretical benchmark — *how close do markets get?*

Competitive markets: setup

Competitive markets for x , z and labor, with prices p_x , p_z and wage w . Person i has income y_i . Firms maximize profits and people maximize utility. No market for E .

Consumers' problem

$$\max_{x_i, z_i} U_i(x_i, z_i, E) + \lambda_i[y_i - p_x x_i - p_z z_i]$$

$$\circ \partial U_i / \partial x_i = \lambda_i p_x, \forall i$$

$$\circ \partial U_i / \partial z_i = \lambda_i p_z, \forall i$$

$$\Rightarrow \frac{\partial U_1 / \partial x_1}{\partial U_1 / \partial z_1} = \frac{p_x}{p_z} = \frac{\partial U_2 / \partial x_2}{\partial U_2 / \partial z_2}$$

Competitive markets: setup

Competitive markets for x , z and labor, with prices p_x , p_z and wage w . Person i has income y_i . Firms maximize profits and people maximize utility. No market for E .

Consumers' problem

$$\max_{x_i, z_i} U_i(x_i, z_i, E) + \lambda_i[y_i - p_x x_i - p_z z_i]$$

- $\partial U_i / \partial x_i = \lambda_i p_x, \forall i$
- $\partial U_i / \partial z_i = \lambda_i p_z, \forall i$

$$\Rightarrow \frac{\partial U_1 / \partial x_1}{\partial U_1 / \partial z_1} = \frac{p_x}{p_z} = \frac{\partial U_2 / \partial x_2}{\partial U_2 / \partial z_2}$$

Firms' problem (for good x)

$$\max_{l_x, E} \{p_x f(l_x, E) - w l_x\}$$

- $p_x(\partial f / \partial l_x) = w$
- $p_x(\partial f / \partial E) = 0$

$\Rightarrow$ firms use E until its marginal product is zero.

Competitive markets: setup

Competitive markets for x , z and labor, with prices p_x , p_z and wage w . Person i has income y_i . Firms maximize profits and people maximize utility. No market for E .

Consumers' problem

$$\max_{x_i, z_i} U_i(x_i, z_i, E) + \lambda_i[y_i - p_x x_i - p_z z_i]$$

- $\circ \partial U_i / \partial x_i = \lambda_i p_x, \forall i$
- $\circ \partial U_i / \partial z_i = \lambda_i p_z, \forall i$

$$\Rightarrow \frac{\partial U_1 / \partial x_1}{\partial U_1 / \partial z_1} = \frac{p_x}{p_z} = \frac{\partial U_2 / \partial x_2}{\partial U_2 / \partial z_2}$$

Firms' problem (for good x)

$$\max_{l_x, E} \{p_x f(l_x, E) - w l_x\}$$

- $\circ p_x(\partial f / \partial l_x) = w$
- $\circ p_x(\partial f / \partial E) = 0$

$\Rightarrow$ firms use E until its marginal product is zero.

Consumers' MRS equals the price ratio, as Pareto optimality requires. The problem is on the firms' side.

Competitive markets: the missing market

Pareto optimality requires

$$-\frac{\partial U_1/\partial E}{\partial U_1/\partial x_1} - \frac{\partial U_2/\partial E}{\partial U_2/\partial x_2} = \frac{\partial f}{\partial E}$$

The marginal product of E equals its marginal damage, summed over people.

Competitive firms choose

$$p_x \frac{\partial f}{\partial E} = 0 \quad \Rightarrow \quad \frac{\partial f}{\partial E} = 0$$

The price of E is zero, so firms use it until its marginal product is zero.

Competitive markets: the missing market

Pareto optimality requires

$$-\frac{\partial U_1/\partial E}{\partial U_1/\partial x_1} - \frac{\partial U_2/\partial E}{\partial U_2/\partial x_2} = \frac{\partial f}{\partial E}$$

The marginal product of E equals its marginal damage, summed over people.

Competitive firms choose

$$p_x \frac{\partial f}{\partial E} = 0 \quad \implies \quad \frac{\partial f}{\partial E} = 0$$

The price of E is zero, so firms use it until its marginal product is zero.

- The two conditions coincide only if $\partial U_i/\partial E = 0$ for everyone, i.e., only if there is no externality.
- Otherwise the competitive allocation is not Pareto optimal.
- The reason is a **missing market**: nobody is paid for accepting emissions and nobody pays for producing them. Markets are not complete, so FWT assumptions are violated.

Competitive markets

Can markets achieve Pareto optimality?

- There's a “**wedge**” between PO allocation and competitive allocation because firms can freely use E as an input to production without accounting for its external costs.
- Governments can intervene by imposing a “**Pigouvian tax**” (named after Arthur Pigou) to be levied on polluting firms equal to the **marginal damages** of each unit of emissions.

$$\text{Set } \tau^* = -p_x \left[\frac{\partial U_1 / \partial E}{\partial U_1 / \partial x_1} + \frac{\partial U_2 / \partial E}{\partial U_2 / \partial x_2} \right]$$

Then, firms maximize, $\max_{l_x, E} \{p_x f(l_x, E) - w l_x - \tau^* E\}$, leading to FOCs:

- $p_x (\partial f / \partial l_x) = w$
- $p_x (\partial f / \partial E) = \tau^* = -p_x \left[\frac{\partial U_1 / \partial E}{\partial U_1 / \partial x_1} + \frac{\partial U_2 / \partial E}{\partial U_2 / \partial x_2} \right]$
 - This is the “**efficiency in emissions**” condition needed for Pareto optimality.

Efficient level of emissions

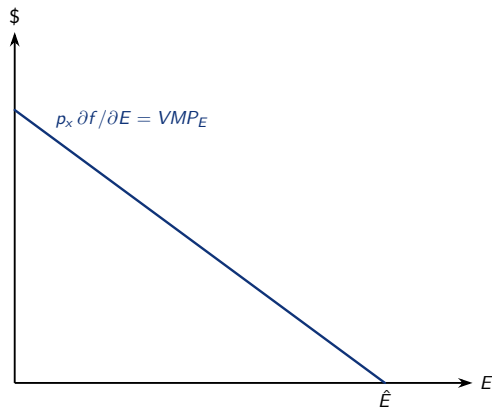


Figure: (P&R-1.1A)

- VMP_E is the "value of marginal product" of E
 - Think of this as a demand (or marginal benefit) curve for emissions.
- Competitive market produces $\hat{E}$ (where $VMP_E = 0$)

Efficient level of emissions

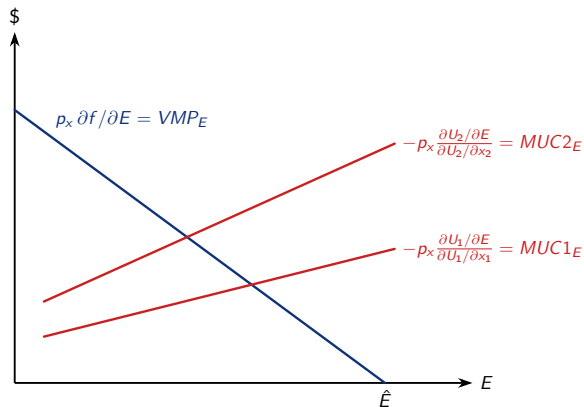


Figure: (P&R-1.1A)

- VMP_E is the "value of marginal product" of E
 - Think of this as a demand (or marginal benefit) curve for emissions.
- Competitive market produces $\hat{E}$ (where $VMP_E = 0$)
- MUC_E is the marginal utility loss of additional emissions.
 - Measured in value of x .

Efficient level of emissions

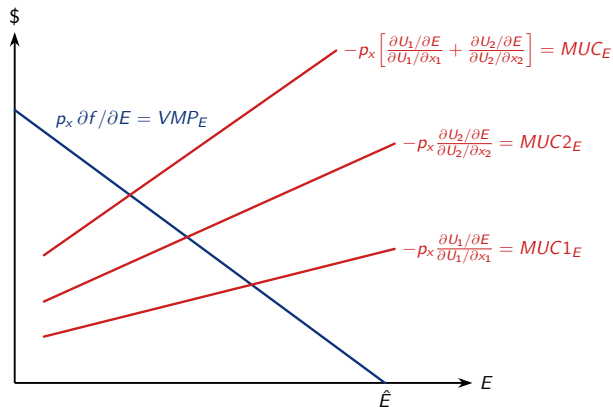


Figure: (P&R-1.1A)

- VMP_E is the "value of marginal product" of E
 - Think of this as a demand (or marginal benefit) curve for emissions.
- Competitive market produces $\hat{E}$ (where $VMP_E = 0$)
- MUC_E is the marginal utility loss of additional emissions.
 - Measured in value of x .
 - Sum of marginal utility loss of all individuals in the economy (here, $N=2$)
 - Can think of this as **marginal external damages**.

Efficient level of emissions

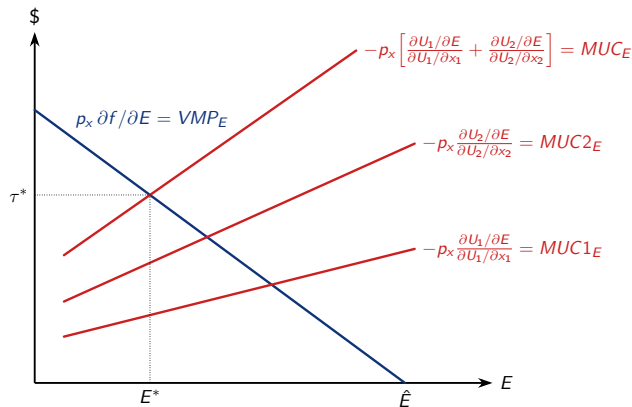


Figure: (P&R-1.1A)

- VMP_E is the "value of marginal product" of E
 - Think of this as a demand (or marginal benefit) curve for emissions.
- Competitive market produces $\hat{E}$ (where $VMP_E = 0$)
- MUC_E is the marginal utility loss of additional emissions.
 - Measured in value of x .
 - Sum of marginal utility loss of all individuals in the economy (here, $N=2$)
 - Can think of this as **marginal external damages**.
- Efficient level of pollution induced by τ^* is E^* , where $E^* < \hat{E}$.

Efficient level of abatement

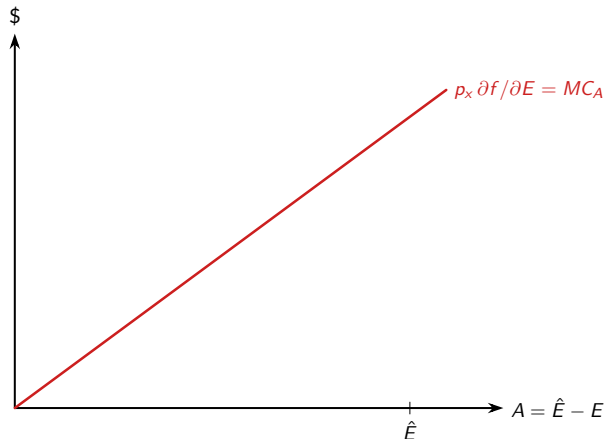


Figure: (P&R-1.1B)

- Emissions are a "bad".
- We can reframe the problem in terms of **pollution abatement**, which is a "good".
 - Flip horizontal axis, $A = \hat{E} - E$ (benefits become costs)
 - MC_A is marg. cost of abatement

Efficient level of abatement

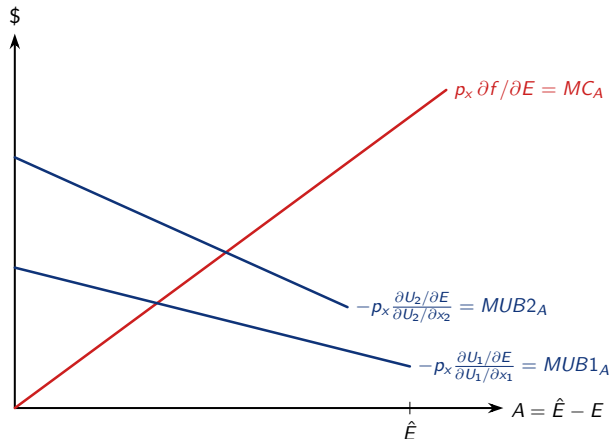


Figure: (P&R-1.1B)

- Emissions are a "bad".
- We can reframe the problem in terms of **pollution abatement**, which is a "good".
 - Flip horizontal axis, $A = \hat{E} - E$ (benefits become costs)
 - MC_A is marg. cost of abatement
- Because pollution abatement is **non-rival** and **non-excludable**, we treat it as a **public good**.

Efficient level of abatement

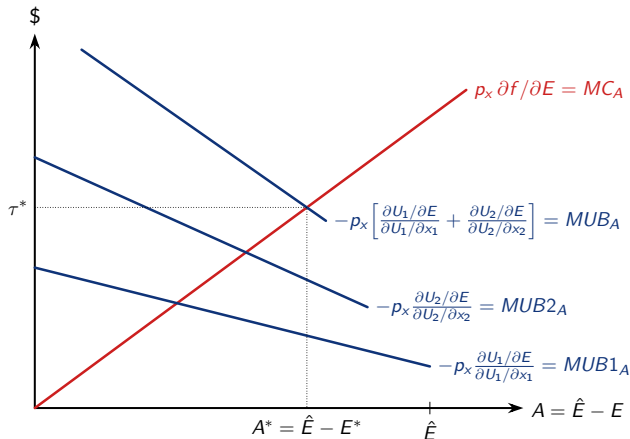


Figure: (P&R-1.1B)

- Emissions are a "bad".
- We can reframe the problem in terms of **pollution abatement**, which is a "good".
 - Flip horizontal axis, $A = \hat{E} - E$ (benefits become costs)
 - MC_A is marg. cost of abatement
 - MUB_A is (collective) marginal WTP for abatement
- Because pollution abatement is **non-rival** and **non-excludable**, we treat it as a **public good**.
 - Vertically sum individual demands to get MUB_A .

Efficient level of abatement

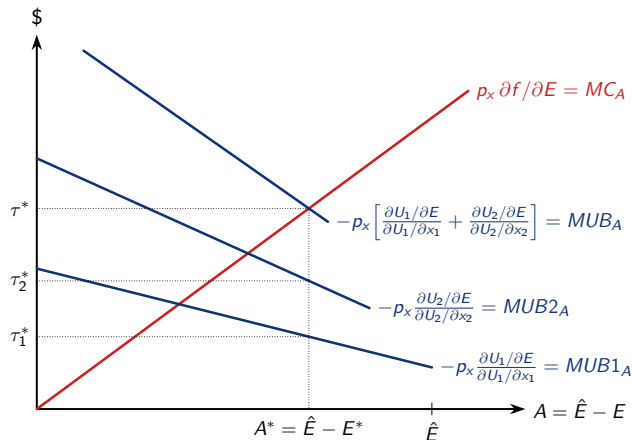
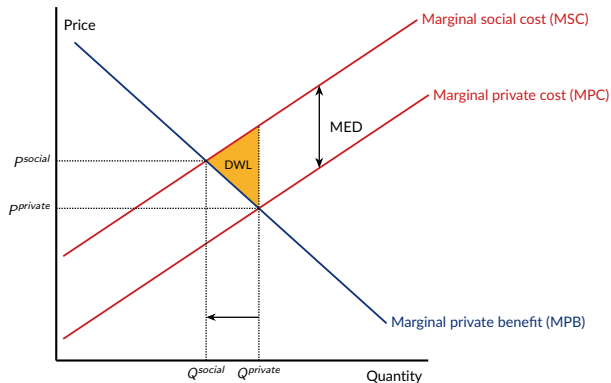


Figure: (P&R-1.1B)

- Emissions are a "bad".
- We can reframe the problem in terms of **pollution abatement**, which is a "good".
 - Flip horizontal axis, $A = \hat{E} - E$ (benefits become costs)
 - MC_A is marg. cost of abatement
 - MUB_A is (collective) marginal WTP for abatement
- Because pollution abatement is **non-rival** and **non-excludable**, we treat it as a **public good**.
 - Vertically sum individual demands to get MUB_A .
 - τ_1^*, τ_2^* are **Lindahl prices**: $\tau_1^* + \tau_2^* = \tau^*$.
 - Competitive markets underprovide public goods because of free riding.

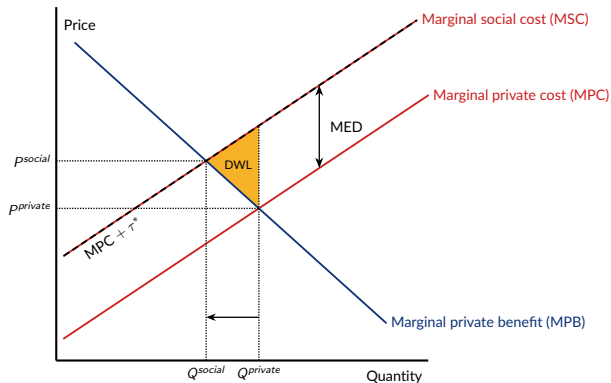
Externalities, summarized

- **Competitive markets**, on their own, will not achieve **Pareto optimality**.
 - The presence of externalities represents a market failure or "missing" market for emissions.
- Firms will produce too much pollution because the price for polluting is $= 0$.



Externalities, summarized

- **Competitive markets**, on their own, will not achieve **Pareto optimality**.
 - The presence of externalities represents a market failure or "missing" market for emissions.
- Firms will produce too much pollution because the price for polluting is $= 0$.
- Governments can intervene by creating a market for pollution (i.e., setting a price equal to **marginal external damages**).
 - We just derived this quantity in the 2-person economy as the sum of marginal disutilities of emissions.
- Firms will internalize this cost and reduce emissions.
- Decentralized, competitive outcome will coincide with the **PO allocation**.



For next class...

caseyjwichman.com/econ7102/#week3

- First reading response due for Coase (1960).
- Due: Tuesday, 11:59 PM.
- Submit via Canvas.